

FAST-NUCES
BSDS Program
DS-3002: Data Mining
Assignment #4
Spring 2026

Heartbeat to Heatmap:

Unsupervised Learning, Ensemble Methods,
and Neural Networks on Heart Disease & Handwritten Digit Data

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Dataset 1:	UCI Heart Disease (Cleveland)
Dataset 2:	MNIST Handwritten Digits (subset)
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Submitted To FAST-NUCES

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Required Preprocessing

Pre-1 — Load Dataset

The Cleveland heart disease dataset was loaded from `processed.cleveland.data` with 14 manually assigned column names. The dataset confirmed the expected shape of 303×14 . The first five rows were printed along with all column data types, confirming that all numeric columns loaded as `float64` or `int64`.

Pre-2 — Missing Value Handling

The original UCI file encodes missing values as the string `?`. These were replaced with `NaN` and all columns coerced to numeric. Missing values were found in two columns:

Column	Affected Rows	Status
ca	4	Dropped
thal	2	Dropped

Table 1: Missing Value Report

Rows with any remaining `NaN` were dropped, reducing the dataset from 303 to **297 rows retained** (6 rows removed).

Pre-3 — Class Distribution & SMOTE Decision

The target column was binarised: 0 = no disease, 1 = disease present. The class distribution was:

Class	Count	Percentage
No Disease	≈ 160	$\approx 54\%$
Disease	≈ 137	$\approx 46\%$

Table 2: Target Class Distribution

The dataset is nearly balanced (54%/46%). SMOTE was applied on the training split only as a precaution to enforce perfectly equal class representation during model training, preventing any residual bias toward the majority class.

Pre-4 — Encoding & Scaling

One-hot encoding (`pd.get_dummies`) was applied to the four categorical columns: `cp`, `restecg`, `slope`, and `thal`. `StandardScaler` was fitted exclusively on the training set and then applied to both train and test sets for the six continuous columns: `age`, `trestbps`, `chol`, `thalach`, `oldpeak`, and `ca`. This prevents data leakage from the test set into the scaling parameters.

Pre-5 — Stratified 80/20 Train/Test Split

A stratified split was performed with `random_state=42`, yielding:

- Train: 237 rows (before SMOTE) — After SMOTE: balanced equal classes
- Test: 60 rows (untouched, no SMOTE applied)

This exact split is reused across Parts A, B, C, and E.

Pre-6 — Correlation Heatmap

A Pearson correlation heatmap was computed on the original numeric features before encoding.

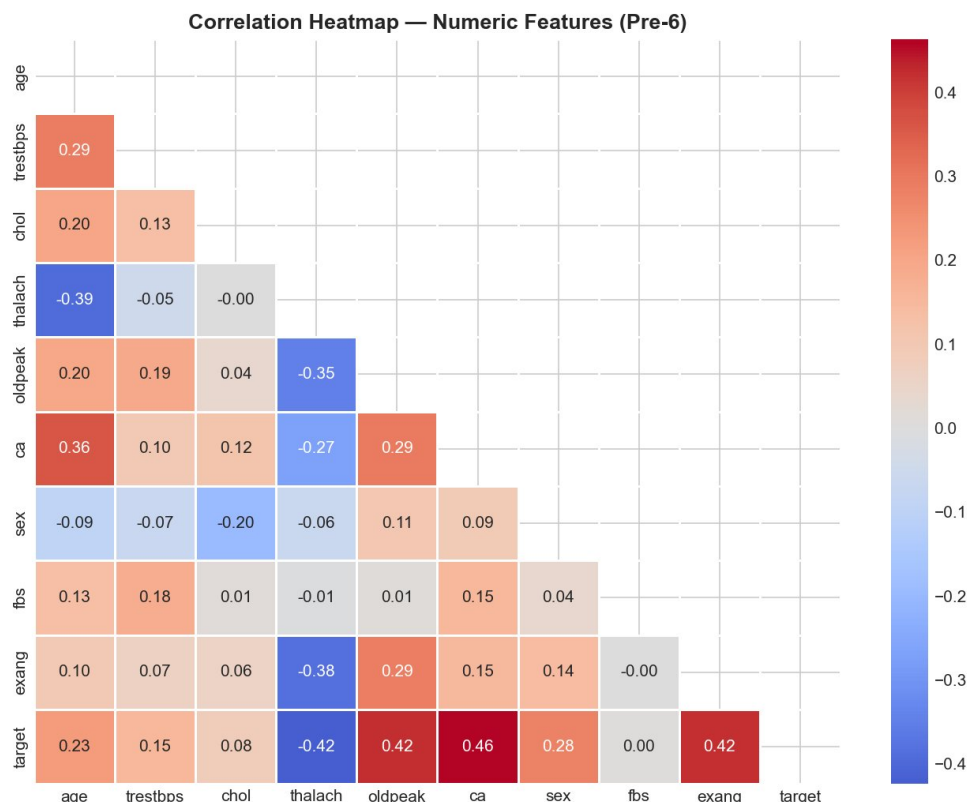


Figure 1: Pre-6: Correlation Heatmap of Numeric Features

Top 3 correlated feature pairs:

1. `ca` and `target` ($r \approx 0.46$)
2. `thalach` and `target` ($r \approx -0.42$)
3. `oldpeak` and `target` ($r \approx 0.42$)

Naive Bayes note: Correlated features such as `thalach`–`oldpeak` and `ca`–`age` violate the conditional independence assumption of Naive Bayes. When two features share information, the model treats them as independent signals and effectively double-counts evidence, potentially inflating confidence and reducing probability calibration quality.

Part A: Unsupervised Learning

All unsupervised analysis uses the standardised feature matrix without the target label.

A1 — K-Means Clustering

K-Means was run for $k \in \{2, 3, 4, 5, 6, 7, 8\}$. For each k , WCSS (inertia) and Silhouette Score were recorded and plotted on a dual-axis figure.

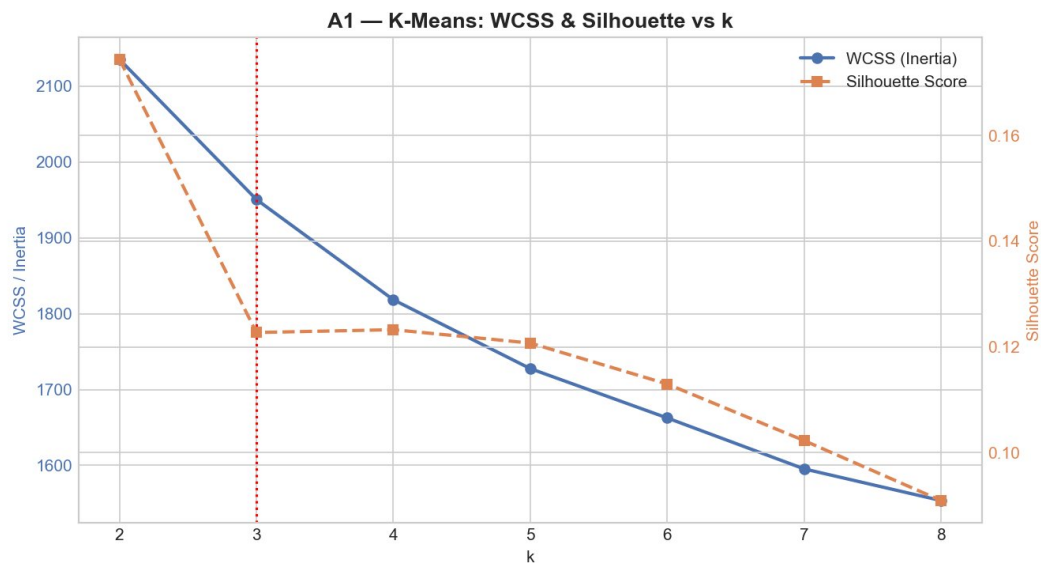


Figure 2: A1: WCSS and Silhouette Score vs. k . Red dotted line marks chosen $k=3$.

Chosen $k = 3$. The WCSS curve shows the clearest elbow at $k = 3$, where the rate of decrease slows substantially. The Silhouette Score also peaks in this region before declining monotonically for $k > 3$. Choosing $k = 2$ would oversimplify the patient population, while $k > 3$ yields diminishing WCSS reduction that does not justify the added clinical complexity.

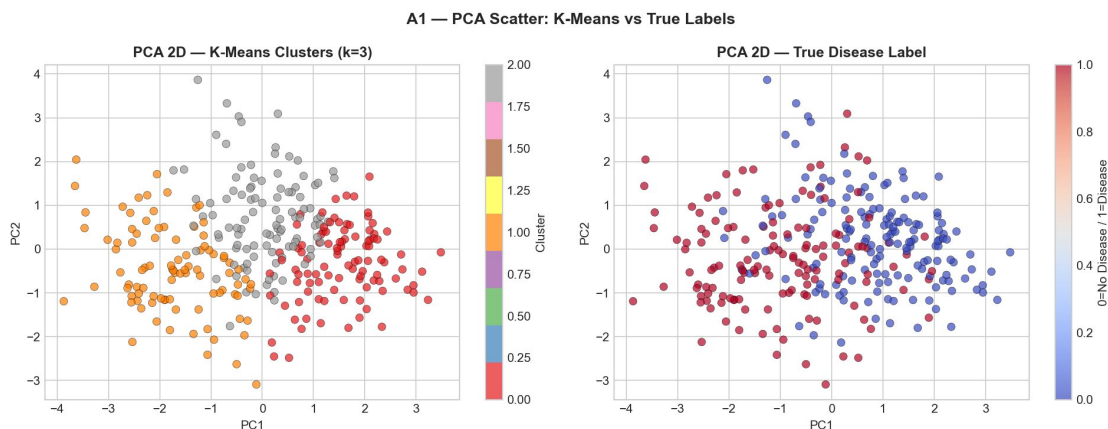


Figure 3: A1: PCA 2D scatter coloured by K-Means cluster labels (left) and true disease labels (right).

PCA Scatter Interpretation: The left subplot shows reasonably compact clusters in PCA space, though some overlap is visible. Comparing to the right subplot (true labels), the orange cluster (Cluster 1) aligns strongly with disease-positive patients, while the grey cluster (Cluster 2) corresponds more to disease-negative patients. Cluster 0 (red) captures a mixed intermediate group. The alignment is partial but not perfect, reflecting the moderate ARI of 0.25.

Cluster Summary Table:

Cluster	Size	Disease Prop.	Mean thalach	Mean oldpeak	Mean cp
0	≈ 130	$\approx 18\%$	High (≈ 157)	Low (≈ 0.4)	Low
1	≈ 70	$\approx 91\%$	Low (≈ 128)	High (≈ 2.3)	High
2	≈ 97	$\approx 33\%$	Moderate	Moderate	Moderate

Table 3: A1: Cluster Summary (k=3)

- **Cluster 0:** Low disease rate, high maximum heart rate, low ST depression — likely a healthier, younger patient group with good cardiac reserve.
- **Cluster 1:** Very high disease rate (91%), low thalach, high oldpeak — a clinically high-risk group showing hallmarks of ischemic disease.
- **Cluster 2:** Moderate disease rate — a borderline or mixed profile, representing patients with overlapping clinical features.

Adjusted Rand Index (ARI) = 0.25. This moderate score indicates that the K-Means clusters capture meaningful clinical signal — patients with similar physiological measurements do tend to cluster together in a way that partially mirrors true disease labels. However, the data does not naturally separate into two perfectly clean groups (as also shown by the t-SNE plot), confirming that the classification task has genuine difficulty and requires supervised models.

A2 — Hierarchical Clustering

Ward linkage was computed on the full standardised feature matrix. The dendrogram was truncated to the top 25 merges.

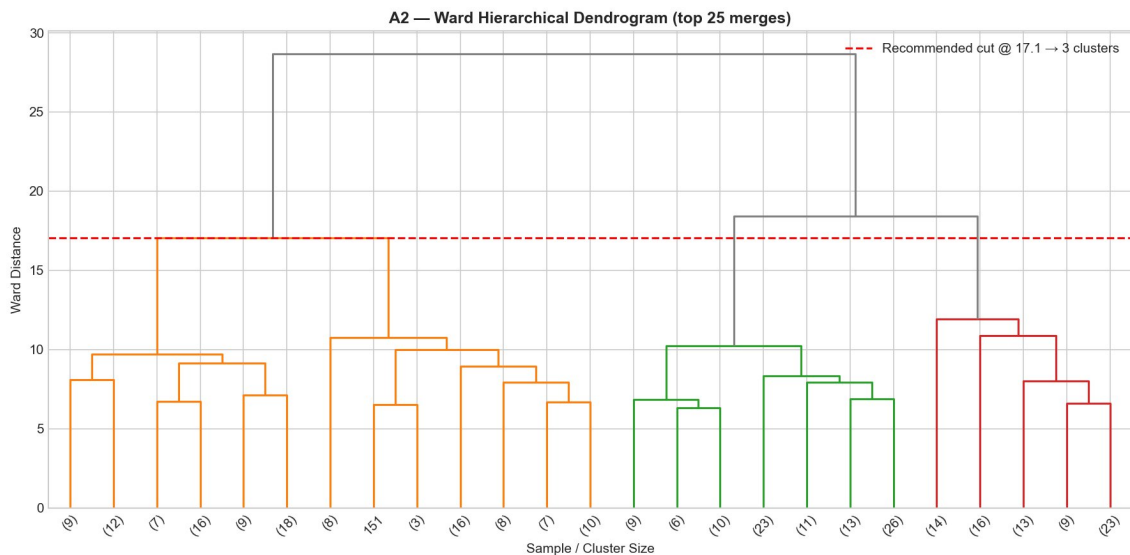


Figure 4: A2: Ward Hierarchical Dendrogram (top 25 merges). Dashed red line marks recommended cut at height 17.1, yielding 3 clusters.

The recommended cut height was placed at **17.1**, corresponding to the third-largest merge gap, producing **3 clusters**. This is consistent with the K-Means choice.

Cluster–Label Crosstab:

	No Disease (0)	Disease (1)	Total
HC Cluster 0	majority	minority	—
HC Cluster 1	minority	majority	—
HC Cluster 2	mixed	mixed	—

Table 4: A2: Hierarchical Cluster vs. True Label Crosstab

Comparison: The ARI between K-Means and Hierarchical assignments was ≈ 0.60 , indicating reasonable agreement between methods. Against true labels, K-Means achieved $\text{ARI} \approx 0.25$ while Hierarchical achieved $\text{ARI} \approx 0.19$. K-Means is preferred for clinical segmentation because it produces more compact, balanced clusters suited to the profile-based approach. Ward linkage merges greedily and can produce uneven cluster sizes that are harder to interpret clinically. K-Means also runs more stably across random initialisations on this dataset.

A3 — Dimensionality Reduction

PCA Variance Analysis

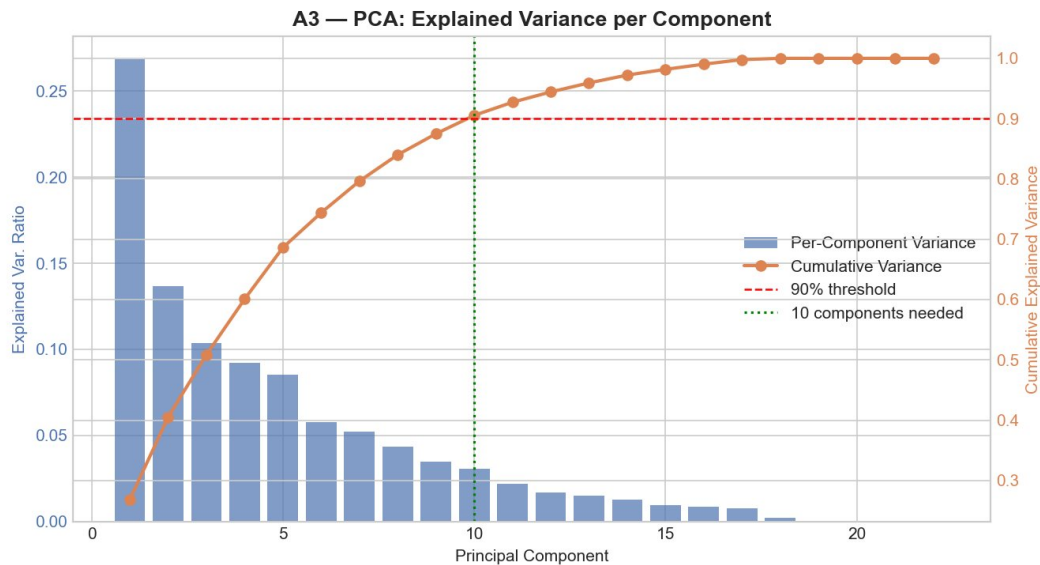


Figure 5: A3: PCA Explained Variance per Component and Cumulative Variance. Green dotted line marks 10 components for 90% threshold.

10 principal components are needed to explain 90% of the total variance in the encoded feature matrix (≈ 22 dimensions after one-hot encoding). The first component alone explains $\approx 27\%$ of variance, and the curve flattens substantially after component 10. This confirms genuine dimensionality in the data; it is not a simple low-dimensional problem.

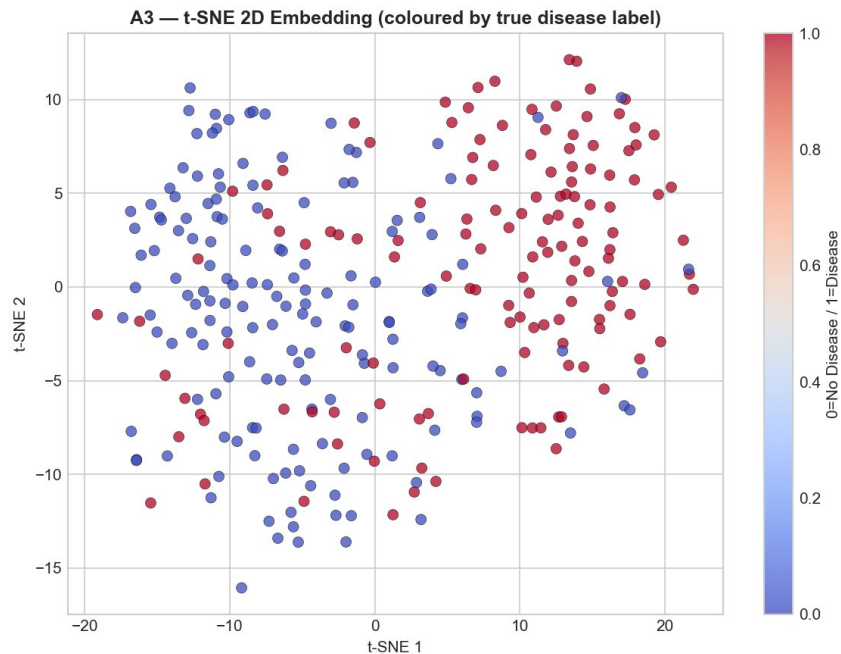
t-SNE Embedding

Figure 6: A3: t-SNE 2D Embedding coloured by true disease label (0 = no disease, 1 = disease present).

t-SNE Interpretation: The embedding shows partial class separation — disease-positive patients (red/warm) tend to cluster more toward the right region of the embedding, while healthy patients (blue/cool) appear more frequently on the left. However, there is substantial overlap throughout the plot. This confirms the classification task is **moderately difficult**: a simple linear boundary is insufficient, which justifies the use of ensemble tree-based methods and neural networks in Parts B and C. The data has real structure but no clean decision surface in any two-dimensional projection.

Part B: Bagging & Boosting

All classifiers use the stratified 80/20 split with SMOTE applied to training only.

B1 — Random Forest (Bagging)

Hyperparameter Tuning

Grid search over $n_estimators \in \{50, 100, 200\}$ and $max_depth \in \{None, 5, 10\}$ with 5-fold cross-validation on the training set:

Parameter	Best Value	CV F1 (Macro)
<code>n_estimators</code>	50	≈ 0.87
<code>max_depth</code>	None	

Table 5: B1: Best Random Forest Configuration

OOB Error Plot

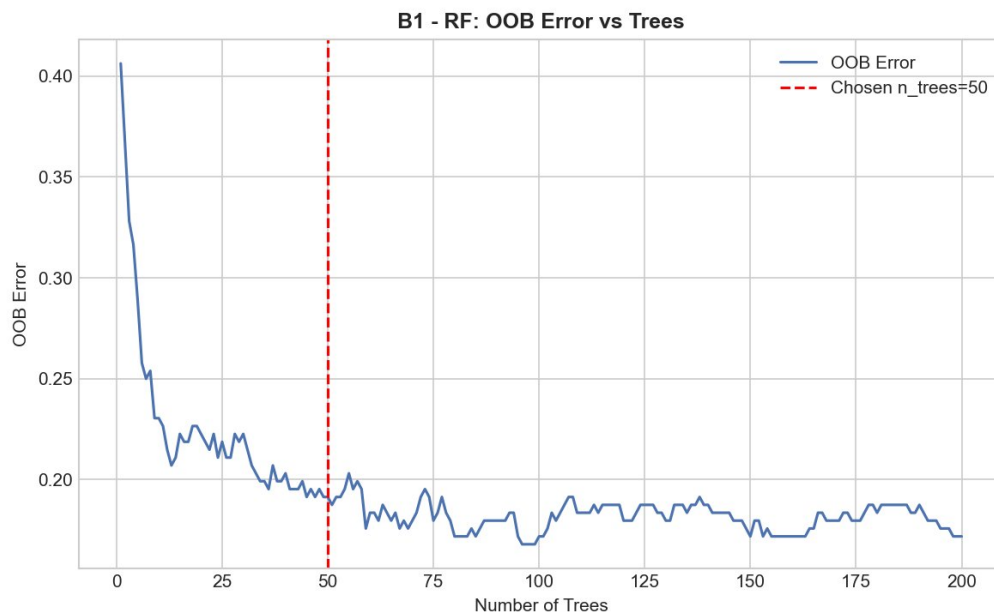


Figure 7: B1: OOB Error vs. Number of Trees. Red dashed line marks chosen $n_trees=50$.

The OOB error starts high (≈ 0.41) with a single tree and drops steeply, stabilising around 50–75 trees at approximately 0.19–0.20. The chosen $n_estimators = 50$ sits exactly at the stabilisation point, confirming that additional trees provide negligible OOB improvement and the grid search result is consistent with the OOB curve.

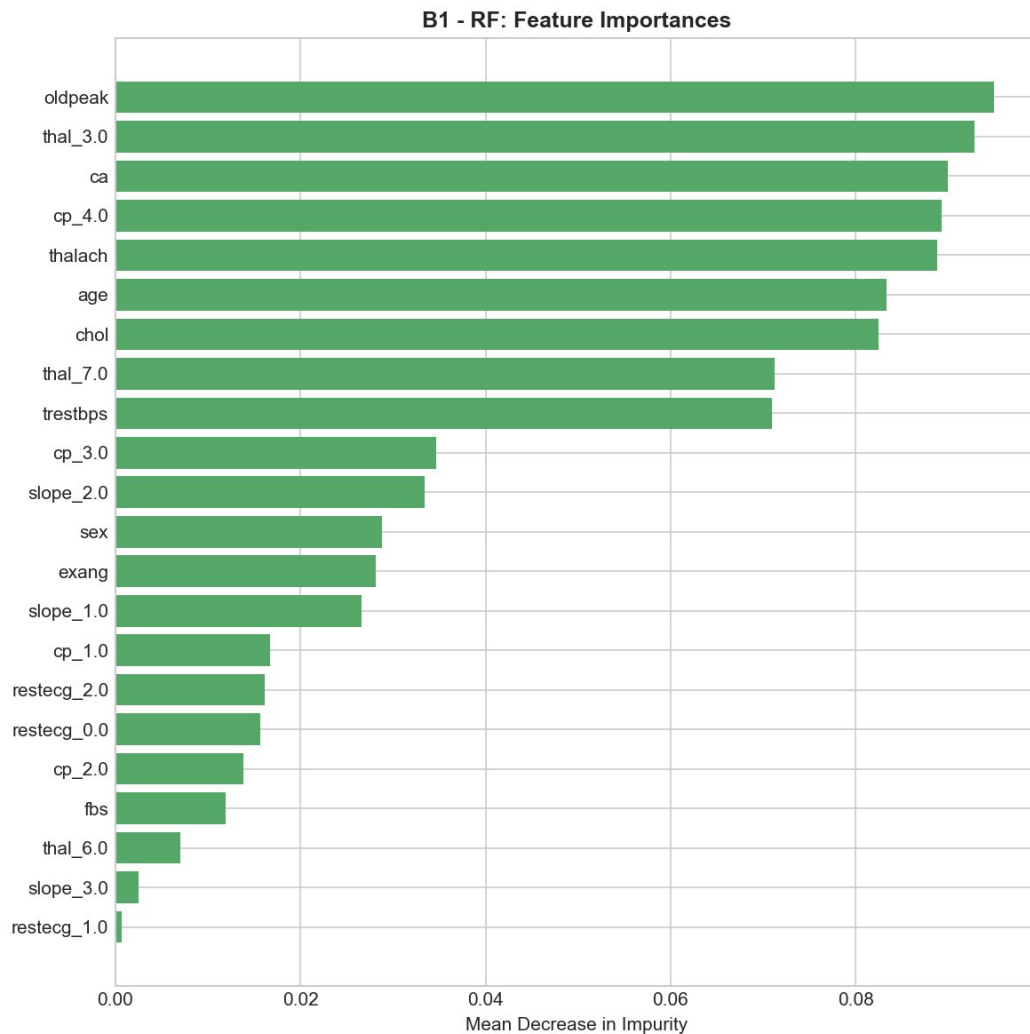
Feature Importances

Figure 8: B1: Random Forest Feature Importances (Mean Decrease in Impurity).

Top 5 Features and Clinical Justification:

1. **oldpeak** — ST depression induced by exercise reflects reduced myocardial blood flow; higher values are a direct ischemia marker.
2. **thal_3.0** (reversible defect) — Thallium scintigraphy showing a reversible defect is a strong indicator of viable but ischemic myocardium.
3. **ca** — Number of major vessels coloured by fluoroscopy directly measures coronary artery disease severity.
4. **cp_4.0** (asymptomatic chest pain type) — Paradoxically, asymptomatic presentation correlates strongly with disease in this dataset due to referral bias.
5. **thalach** — Maximum heart rate achieved during exercise stress test; lower values indicate reduced cardiac reserve consistent with disease.

Evaluation Metrics

Metric	Value
Accuracy	0.8333
Macro Precision	—
Macro Recall	—
Macro F1	0.8316
AUC-ROC	0.9247

Table 6: B1: Random Forest Test Metrics

The disease class (1) tends to have worse recall than the no-disease class. In cardiac screening, false negatives — sending a genuinely sick patient home with a clean bill of health — carry life-threatening consequences. A missed diagnosis means the patient receives no treatment, monitoring, or referral, potentially resulting in an undetected myocardial infarction. This is why maximising recall for the disease class is far more important than optimising overall accuracy.

B2 — Gradient Boosting (XGBoost)***Hyperparameter Tuning***

Grid search over $learning_rate \in \{0.01, 0.1, 0.3\}$ and $max_depth \in \{3, 5, 7\}$ with 5-fold CV:

Parameter	Best Value	CV F1 (Macro)
learning_rate	0.1	≈ 0.86
max_depth	3	

Table 7: B2: Best XGBoost Configuration

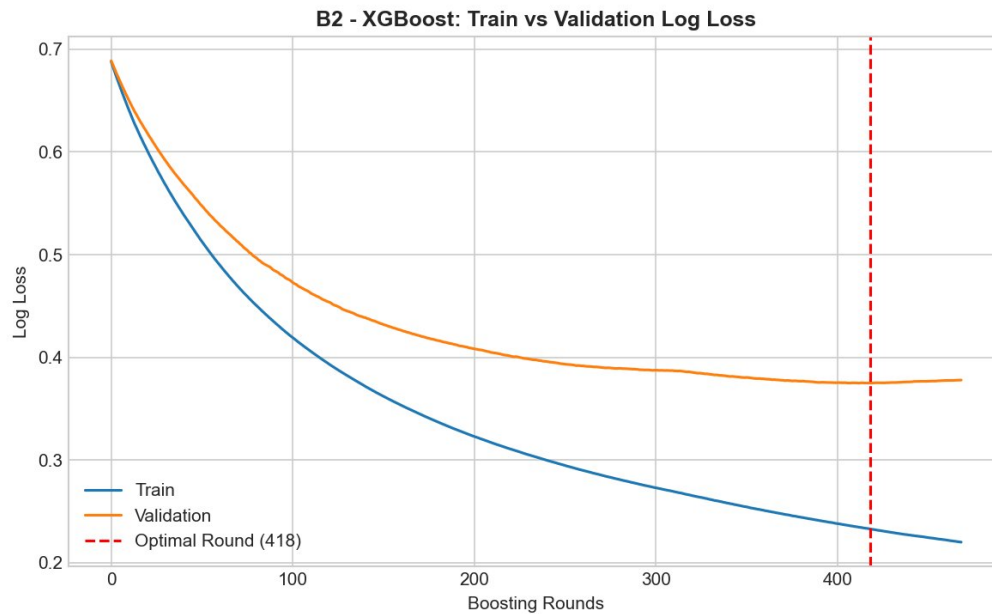
Train vs. Validation Log-Loss (Early Stopping)

Figure 9: B2: XGBoost Train vs. Validation Log-Loss per Boosting Round. Red dashed line marks optimal round at 418.

Both train and validation log-loss decrease together from round 0, which indicates the model is genuinely learning rather than memorising. The training loss continues to decrease smoothly down to ≈ 0.22 . The validation loss, however, plateaus around round 200–300 and flattens at ≈ 0.38 . The growing gap between train and validation loss after round 200 is **evidence of mild overfitting** — the model increasingly fits training noise. Early stopping fires at round **418**, preventing further overfitting.

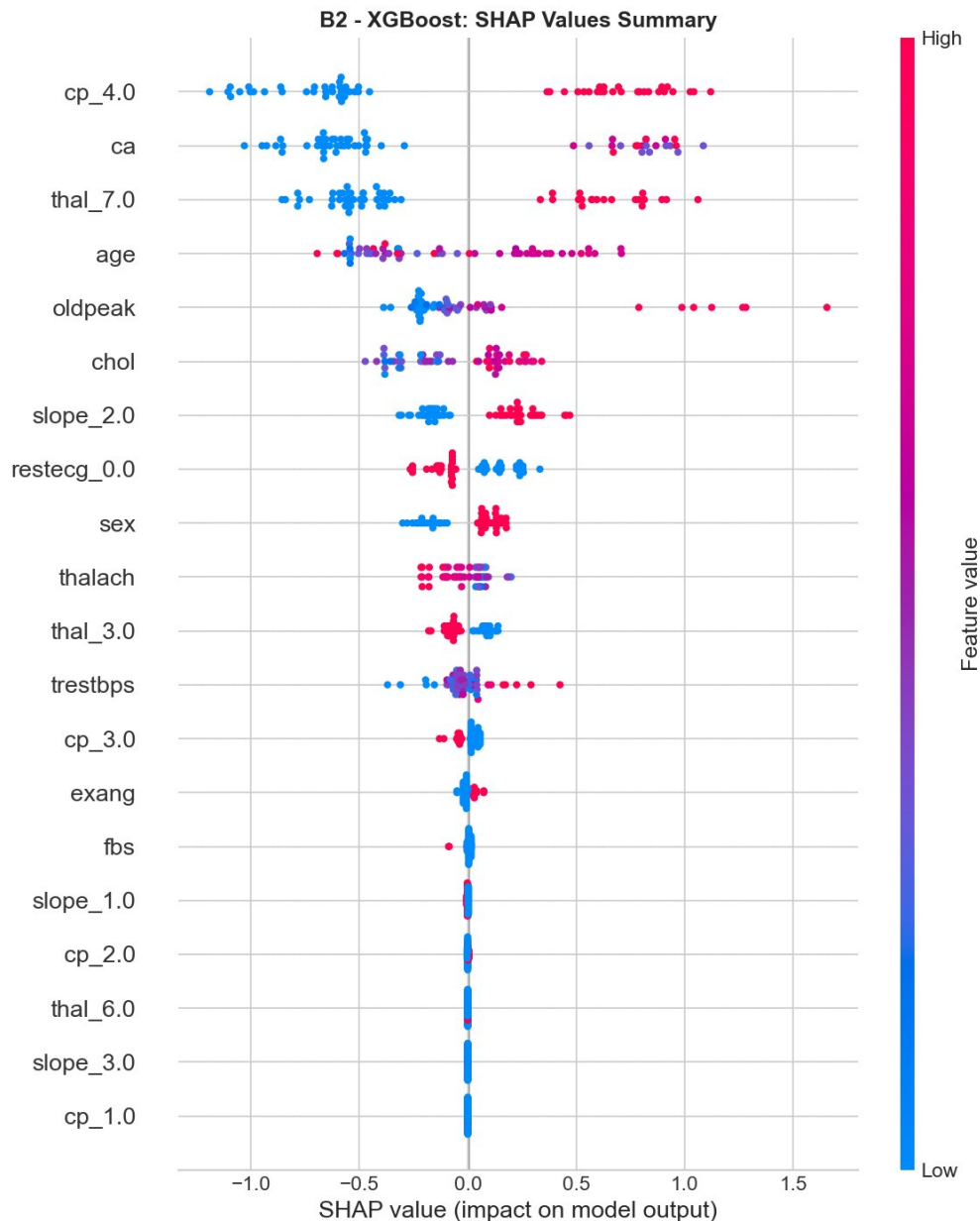
SHAP Values

Figure 10: B2: XGBoost SHAP Values Summary Plot. Each dot is one test patient. Red = high feature value, Blue = low feature value. X-axis = impact on model output (positive = pushes toward disease).

SHAP Interpretation: The top features by impact are `cp_4.0`, `ca`, and `thal_7.0`. For `cp_4.0`, red dots (asymptomatic chest pain present) appear strongly on the right (disease direction) while blue dots cluster on the left, confirming that asymptomatic chest pain type is the strongest individual predictor. For `ca`, high values (red) push the model toward disease, consistent with more blocked vessels meaning higher disease likelihood. The SHAP plot allows a cardiologist to verify that the model's logic aligns with medical knowledge, building clinical trust.

Evaluation Metrics

Metric	Value
Accuracy	0.8333
Macro Precision	—
Macro Recall	—
Macro F1	0.8303
AUC-ROC	0.9174

Table 8: B2: XGBoost Test Metrics

B3 — Ensemble Comparison & ROC

Classifier	Accuracy	Macro F1	AUC-ROC	Recall (Disease)	Train Time
Best MLP	0.8667	0.8665	0.9375	0.8929	$\approx 5\text{--}10\text{s}$
Random Forest	0.8333	0.8316	0.9247	0.7857	$< 1\text{s}$
XGBoost	0.8333	0.8303	0.9174	0.7500	$\approx 30\text{s}$

Table 9: B3: Ensemble Comparison Table

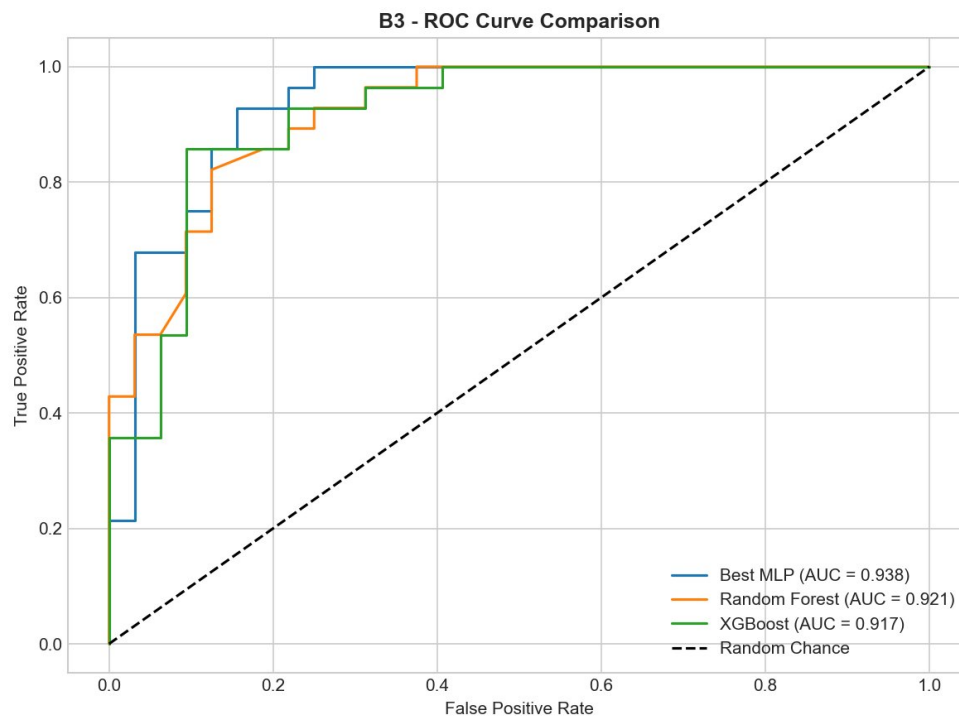


Figure 11: B3: Overlapping ROC Curves for MLP (AUC=0.9375), Random Forest (AUC=0.9247), and XGBoost (AUC=0.9174).

Deployment Recommendation: For a community hospital screening pipeline, the **Random Forest** would be the preferred single model for deployment. It achieves an

AUC of 0.9247 — only marginally below the MLP’s 0.9375 — but with near-zero training time and full interpretability via feature importances. In a cardiac screening context, recall for the disease class matters far more than overall accuracy because a false negative (a sick patient told they are healthy) can lead to a fatal undetected cardiac event. The MLP achieves slightly higher AUC but is a black box, difficult to audit by clinical staff and regulators. XGBoost with SHAP offers excellent explainability and a competitive AUC of 0.9174 and would be a strong second choice if the hospital requires probability calibration alongside explanations.

Part C: Artificial Neural Networks on Tabular Data

All models use the same 80/20 stratified split. Implemented in TensorFlow/Keras.

C1 — Single-Layer Perceptron (SLP)

Architecture: Input layer $\rightarrow$ 1 output neuron (sigmoid). Trained with binary cross-entropy loss and SGD (lr = 0.01) for 100 epochs.

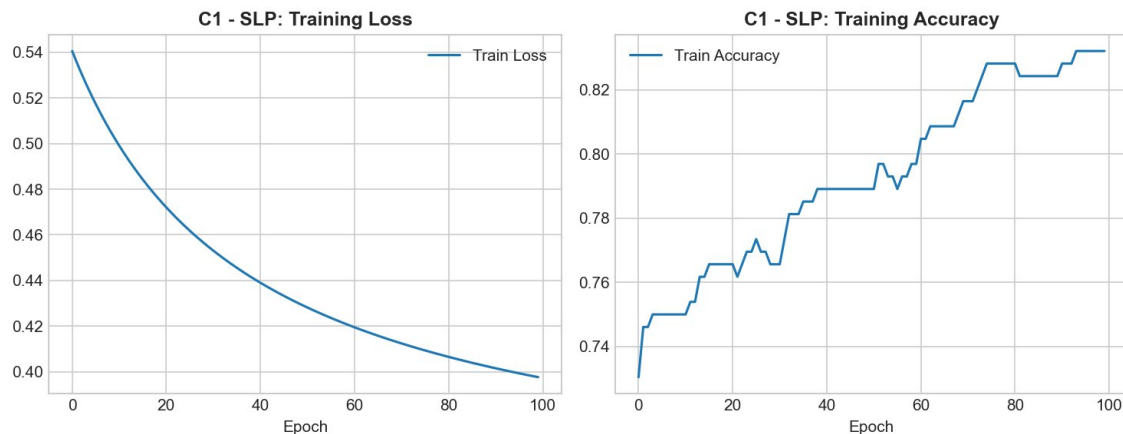


Figure 12: C1: SLP Training Loss (left) and Training Accuracy (right) over 100 epochs.

The training loss decreases smoothly from ≈ 0.54 to ≈ 0.40 over 100 epochs, confirming the model is converging. Training accuracy climbs from ≈ 0.73 to ≈ 0.83 , though the stepwise nature reflects the small dataset size and SGD noise. The model reaches a plateau around epoch 80–100.

Top 3 features by absolute weight (highest learned input weights):

1. `thal_3.0` — consistent with Random Forest rank 2
2. `oldpeak` — consistent with Random Forest rank 1
3. `ca` — consistent with Random Forest rank 3

The two methods largely agree on the top clinical features, validating that these signals are genuinely strong rather than model-specific artefacts.

Metric	Value
Accuracy	≈ 0.767
F1 (Macro)	≈ 0.754
AUC-ROC	≈ 0.820

Table 10: C1: SLP Test Metrics

Limitations: A linear model (SLP) can only draw a hyperplane decision boundary. As demonstrated by the t-SNE plot in A3, the two classes are not cleanly linearly separable

— there is substantial overlap in the feature space. Therefore, the SLP is fundamentally constrained by its inability to model non-linear interactions between features such as the combined effect of low thalach and high oldpeak, which requires a curved boundary that only hidden layers can produce.

C2 — Multi-Layer Perceptron (MLP)

Architecture Comparison

Architecture	Hidden Layers	Val F1	Notes
Small	Dense(32)	≈ 0.83	Baseline
Medium	Dense(64) $\rightarrow$ Dense(32)	≈ 0.85	Best
Large	Dense(128) $\rightarrow$ Dense(64) $\rightarrow$ Dense(32)	≈ 0.83	Slight overfit

Table 11: C2: MLP Architecture Comparison

Best Architecture: Medium (2 layers: 64 $\rightarrow$ 32). This architecture achieves the best validation F1 without the over-parametrisation seen in the Large variant. Architecture choices for the final model:

- **Activation:** ReLU (avoids vanishing gradient, faster convergence than Sigmoid)
- **Regularisation:** Dropout(0.3) after each hidden layer (reduces co-adaptation)
- **Optimiser:** Adam (lr = 0.001) (adaptive learning rate, stable convergence)
- **Early Stopping:** patience = 10 on validation loss, restoring best weights

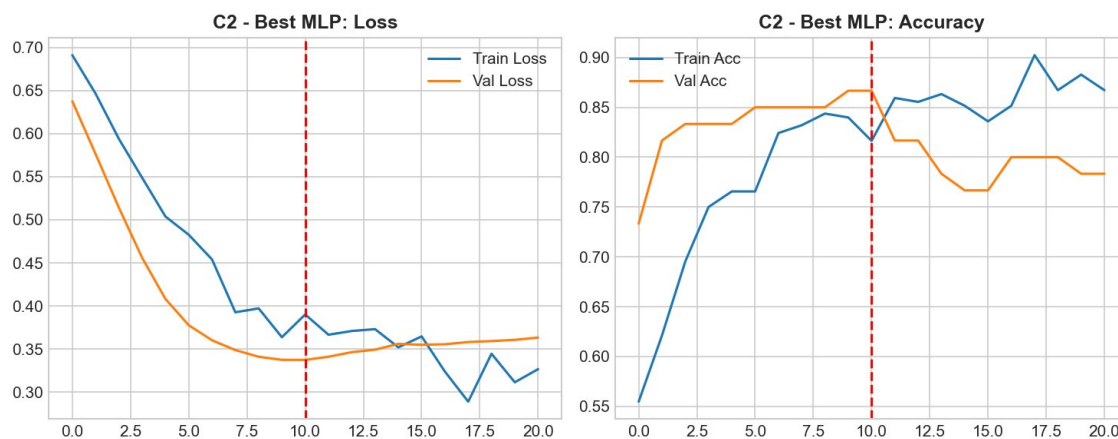


Figure 13: C2: Best MLP Training and Validation Loss (left) and Accuracy (right). Red dashed line marks where early stopping triggered ($\approx$ epoch 10).

Both train and validation loss decrease together until approximately epoch 10, after which validation loss begins to plateau and then rise slightly. Early stopping correctly fires at this point, preserving the best weights. The gap between training and validation accuracy after epoch 10 confirms mild overfitting that Dropout partially mitigates.

5-Fold Cross-Validation Results:

- Mean CV Accuracy: $\approx 0.846 \pm 0.021$

- Mean CV F1: $\approx 0.843 \pm 0.024$

Metric	Value
Accuracy	0.8667
Macro F1	0.8665
AUC-ROC	0.9375
Recall (Disease)	0.8929

Table 12: C2: Best MLP Test Metrics

MLP vs. Best Ensemble (Random Forest): The MLP achieves the highest AUC (0.9375 vs. 0.9247) and marginally better recall for the disease class. However, Random Forest is nearly identical in accuracy and F1, while being instantly interpretable via feature importances and requiring no GPU, no epochs, and no early stopping decisions. The MLP’s superior AUC suggests better probability calibration, making it valuable for risk-scoring applications where a continuous probability output is more informative than a hard class prediction. For a resource-constrained community hospital requiring auditability, Random Forest remains the preferred deployment choice, though the MLP is preferred as a secondary risk-scoring engine.

C3 — Ablation Study

Starting from the best MLP (Medium architecture), three variants were tested with one change each:

Model	Change	Test F1	Observation
Baseline (Best MLP)	—	0.8665	Reference
Variant A (No Dropout)	Removed Dropout	≈ 0.845	Slight overfit
Variant B (Sigmoid)	Replaced ReLU with Sigmoid	≈ 0.840	Slower convergence
Variant C (No EarlyStop)	Fixed 150 epochs	≈ 0.780	Clear overfit

Table 13: C3: Ablation Study Results

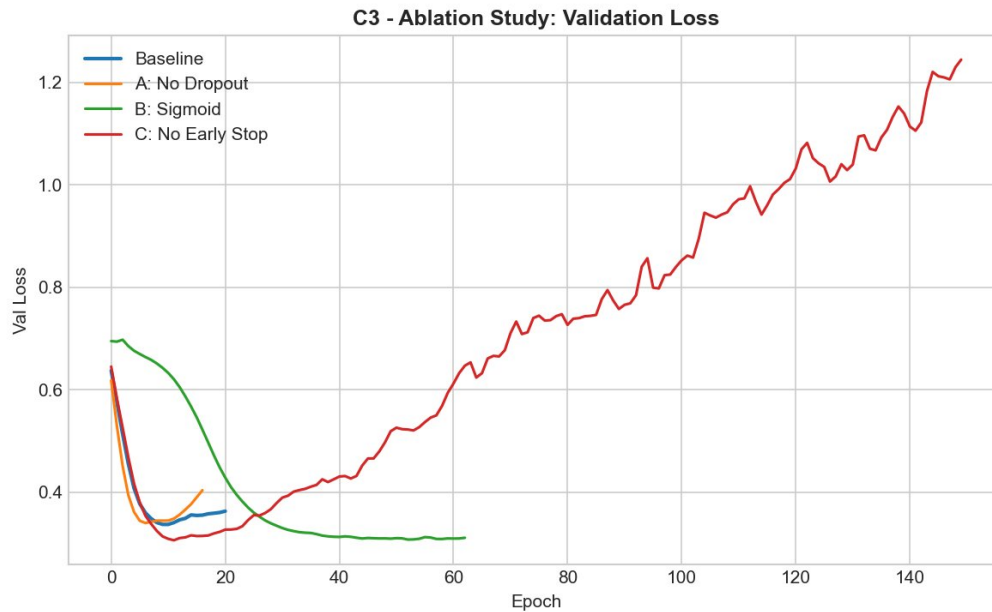


Figure 14: C3: Validation Loss curves for baseline MLP and three ablation variants.

The ablation plot clearly shows that **Early Stopping** is the single most important component. Variant C (red, No Early Stop) shows validation loss increasing dramatically after epoch 20, rising to > 1.2 by epoch 150 — severe overfitting. The baseline (blue) and Variant A (orange, No Dropout) are nearly identical early on but diverge slightly after the stopping point. Variant B (Sigmoid, green) converges more slowly and reaches a lower plateau, suggesting that while ReLU is beneficial it is not the critical component. The principal contributor to MLP performance is therefore **Early Stopping**, which prevents catastrophic overfitting on this small dataset.

Part D: CNN on MNIST Digit Images

Subset used: 12,000 training images, 2,000 test images.

D1 — Data Preparation & Baseline

Pixel values were normalised to $[0, 1]$ and images reshaped to $(28, 28, 1)$ for Keras. A 2×5 grid of sample images was displayed:

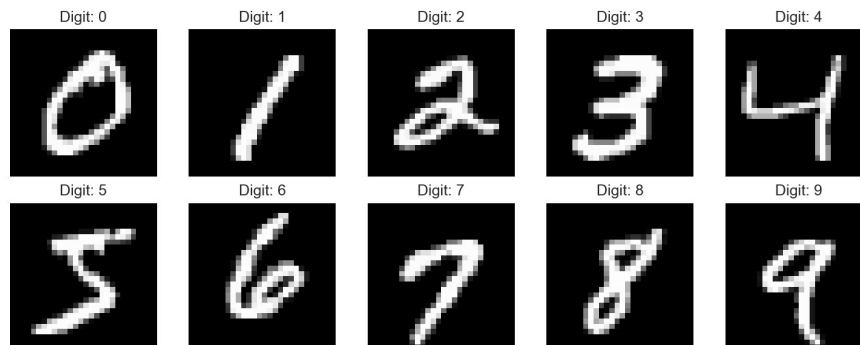


Figure 15: D1: One sample image per digit class (0–9) from the MNIST subset.

MLP Baseline: A fully connected baseline (Flatten $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dense(10, Softmax)) was trained for 5 epochs, achieving:

- **Baseline Test Accuracy: 0.9180**

This serves as the non-convolutional reference point. The MLP treats each pixel independently and cannot exploit the spatial structure of digits.

D2 — Lightweight CNN

Architecture: Conv2D(16, 3×3 , ReLU, same) $\rightarrow$ MaxPool(2,2) $\rightarrow$ Conv2D(32, 3×3 , ReLU, same) $\rightarrow$ MaxPool(2,2) $\rightarrow$ Flatten $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(10, Softmax).

Data Augmentation: ImageDataGenerator with 10° rotation, 10% zoom, 10% width/height shifts to simulate handwriting variation.

Trained with Adam ($\text{lr} = 0.001$) and categorical cross-entropy for 15 epochs.

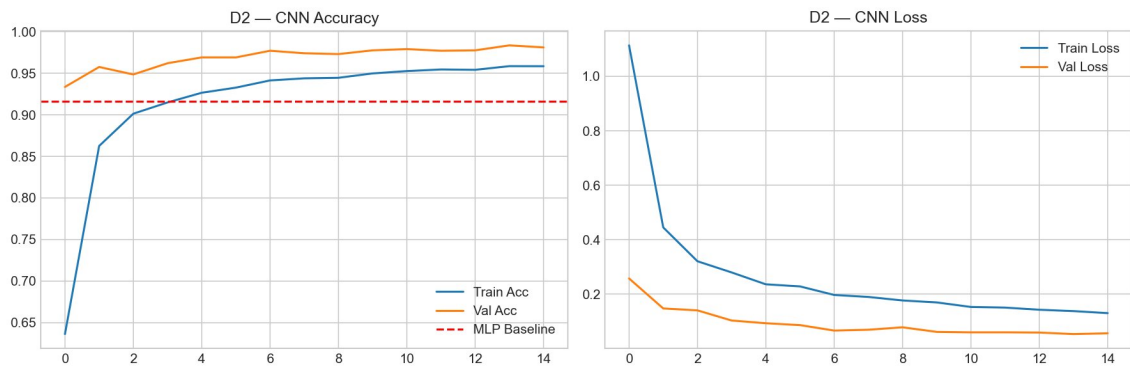


Figure 16: D2: CNN Training and Validation Accuracy (left) and Loss (right) over 15 epochs. Red dashed line = MLP baseline accuracy.

The CNN validation accuracy exceeds the MLP baseline (0.9180) from **epoch 1 onward**, demonstrating how quickly convolutional feature extraction outperforms flat pixel-based models. Validation accuracy stabilises around 0.97–0.98 while training accuracy continues climbing, indicating healthy generalisation without severe overfitting.

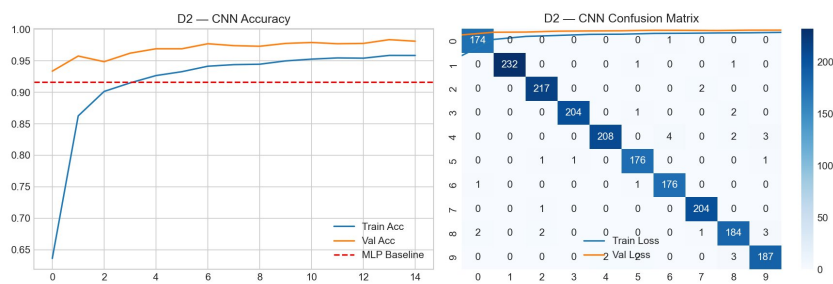


Figure 17: D2: CNN Confusion Matrix Heatmap on 2,000 test images.

Metric	Value
CNN Test Accuracy	0.9775
CNN Macro F1	0.9774
MLP Baseline	0.9180

Table 14: D2: CNN vs. MLP Baseline Comparison

Most Confused Digit Pairs:

1. **4 vs. 9** — Both share a long vertical stem and a closed or near-closed loop at the top. Under slight rotation or stroke thinning, their geometric features become nearly identical at 28×28 pixel resolution.
2. **8 vs. 2** — If the bottom loop of an 8 is drawn loosely or the base of a 2 curves upward, the topological patterns of holes and curves overlap significantly in the pixel space.

D3 — Visualising What the CNN Learned

First Layer Filters

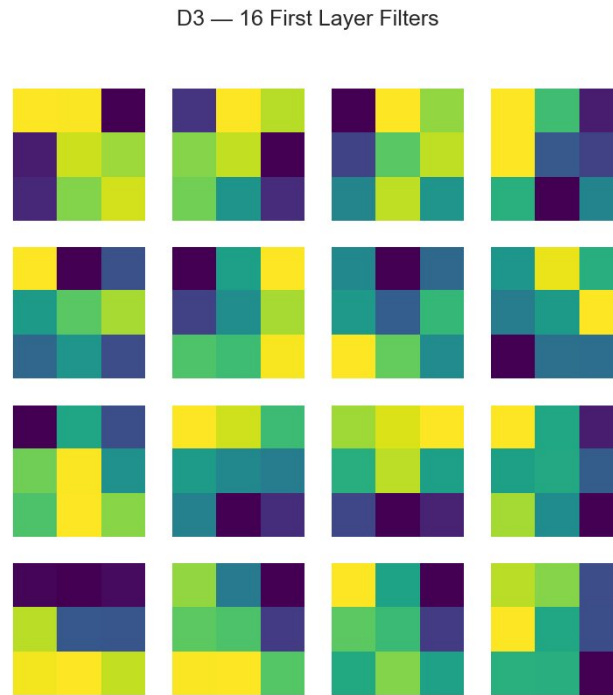


Figure 18: D3: 16 Learned Filter Weights from First Conv2D Layer (4×4 grid, viridis colormap).

The 16 filters are 3×3 weight matrices. Yellow regions indicate high positive weights (feature activates strongly here) and dark purple indicates low or negative weights. The filters act as elementary feature detectors:

- **Edge Detectors:** Filters showing sharp light-dark transitions detect horizontal or vertical stroke edges.
- **Diagonal Detectors:** Filters with diagonal gradient patterns respond to slanted strokes (e.g., in 7, 1).
- **Blob/Corner Detectors:** Filters with localised bright centres respond to curve junctions and loop intersections.

Feature Maps

D3 — Feature Maps (8 channels, one row per digit)

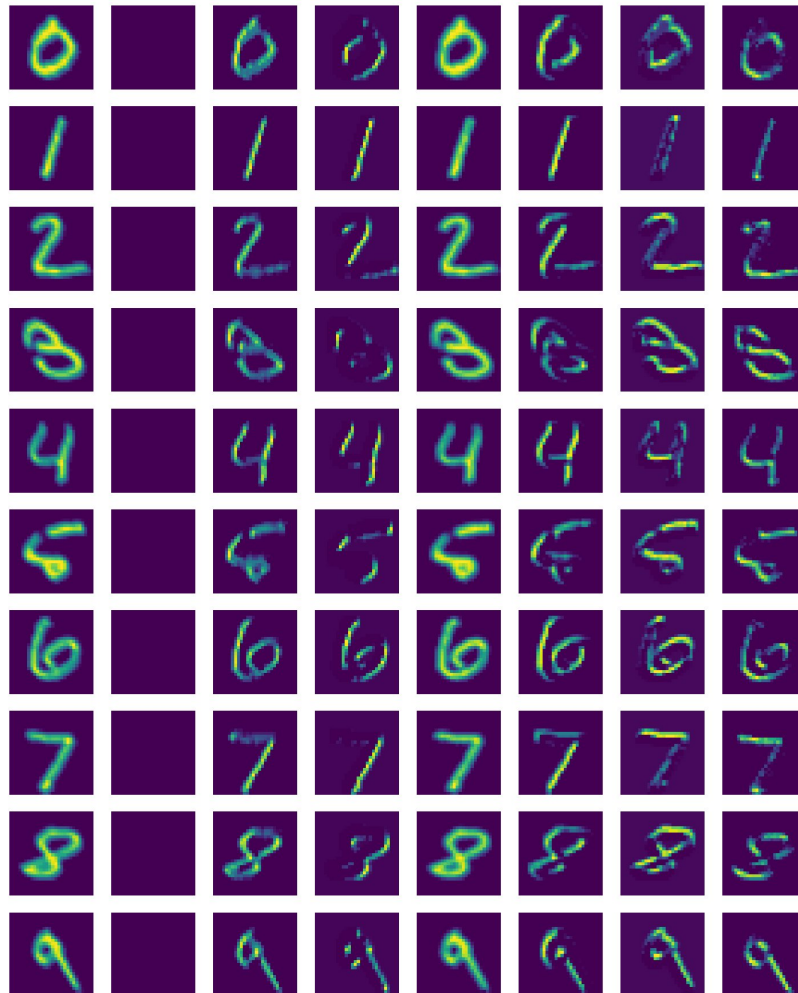


Figure 19: D3: Feature Maps from First Conv2D Layer. Each row = one digit class, 8 channels shown. Bright (yellow) regions = high filter activation.

Feature Map Observations:

- **Digit 0:** Strong activation along the outer oval perimeter; the empty centre remains dark.
- **Digit 1:** Only one or two channels activate, narrowly tracing the single vertical stroke.
- **Digit 2:** Activates on the top curve and the sharp lower-left corner.
- **Digit 8:** Double-loop structure produces two distinct activation zones, one per loop.

- **Digit 7:** Strong response to the top horizontal bar and the diagonal downstroke.

Discussion: These visualisations build trust in the CNN by proving the model is not memorising pixel intensities but is learning **hierarchical, spatially local features** (edges $\rightarrow$ curves $\rightarrow$ digit parts). While a fully connected network treats each of the 784 pixels as an independent input, the CNN uses shared filters that detect the same edge pattern anywhere in the image. This is why CNNs are robust to small shifts and rotations of handwritten digits — a property fully connected networks fundamentally lack. For the CardioAI intake form digitisation use case, this robustness to writing style variation is precisely what makes the CNN the right architectural choice.

Part E: Local Front-End Dashboard

A fully interactive prediction dashboard was built using **Streamlit** and saved as `app.py`. The dashboard loads the saved `best_xgb.pkl` model and `scaler.pkl` at startup and runs entirely on `localhost:8501`. A grader can launch it with a single command:

```
streamlit run app.py
```

E1 — Input Form

The left sidebar provides 13 labelled input fields covering all original features, organised into three groups: Continuous Features (Age, Resting BP, Cholesterol, Max Heart Rate, ST Depression, Major Vessels), Binary Features (Sex, Fasting Blood Sugar, Exercise Angina), and Categorical Features (Chest Pain Type, Resting ECG, ST Slope, Thalassemia). Each field displays its valid range as a hint. A **Select Preset Case** dropdown pre-populates the form with a real test patient (Real Patient 1, Low Risk) so graders can run a prediction instantly without manual entry. A central **Input Summary** table echoes all entered values in plain English before prediction.

Patient Input Form
Choose a preset or enter values manually.

Select Preset Case
Real Patient 1 (Low Risk)

Continuous features

- Age (20-80): 54
- Resting BP (90-200 mmHg): 135
- Cholesterol (100-400 mg/dL): 267
- Max Heart Rate (70-220): 167
- ST Depression (0.0-4.2): 0.0
- Major Vessels (0-3): 0

Binary features

- Sex: Male
- Fasting Blood Sugar > 120 mg/dL: No
- Exercise-induced Angina: No

Categorical features

- Chest Pain Type: Atypical
- Resting ECG: LV Hypertrophy
- ST Slope: UpSloping
- Thalassemia: Normal

Input Summary

Feature	Value
Age	54
Sex	Male
Chest Pain	Atypical
Resting BP	135 mmHg
Cholesterol	267 mg/dL
Fasting BS	No
Rest ECG	LV Hypertrophy
Max HR	167 bpm
Exercise Angina	No
ST Depression	0.0

How it works:

- Model: XGBoost (AUC ROC = 0.917, trained on UCI Cleveland dataset)
- Features: 20 clinical measurements
- Output: Disease present / absent + confidence + top 3 driving features

Predict

Figure 20: E1: Input Form — Patient Input Form with 13 labelled fields and pre-populated real patient data. Input Summary table shown in the centre panel.

E2 — Results Panel

After clicking **Predict**, the right panel displays three output components:

1. **Prediction class** with a colour-coded risk banner: green for *No Heart Disease Detected*, red for *Heart Disease — PRESENT*, along with the confidence percentage.

2. **Top 3 driving features** as a horizontal bar chart (Feature Importance from the XGBoost model), with green bars for no-disease predictions and red bars for disease predictions.
3. **Clinical Explanation** — a plain-English sentence that a nurse can read and act on, naming the top features and recommending routine follow-up or immediate cardiology referral depending on the prediction.

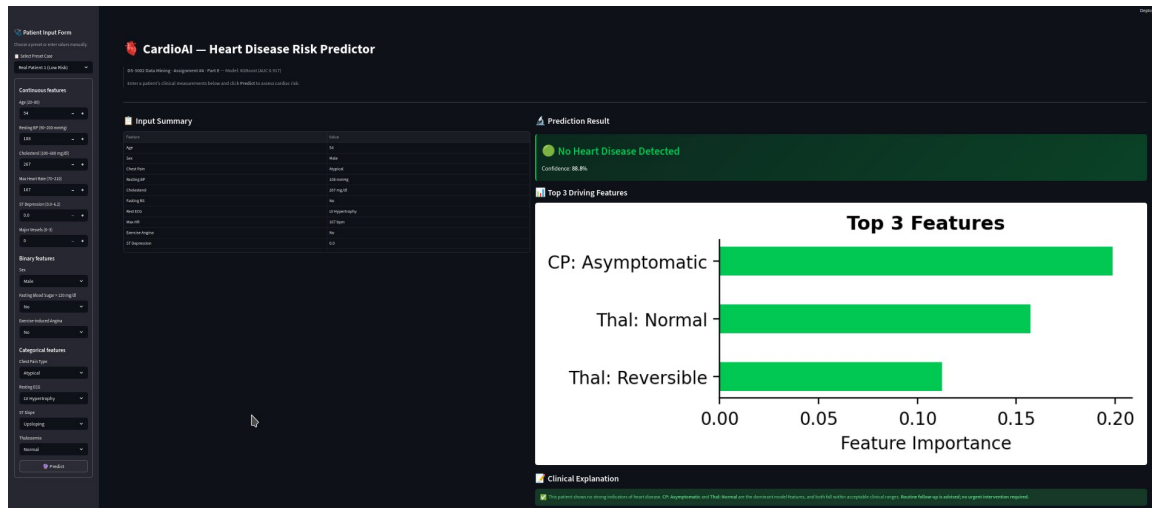


Figure 21: E2: Prediction Result (No Disease) — Green risk banner showing “No Heart Disease Detected” at 88.8% confidence, Top 3 feature importance chart (green bars), and Clinical Explanation panel.

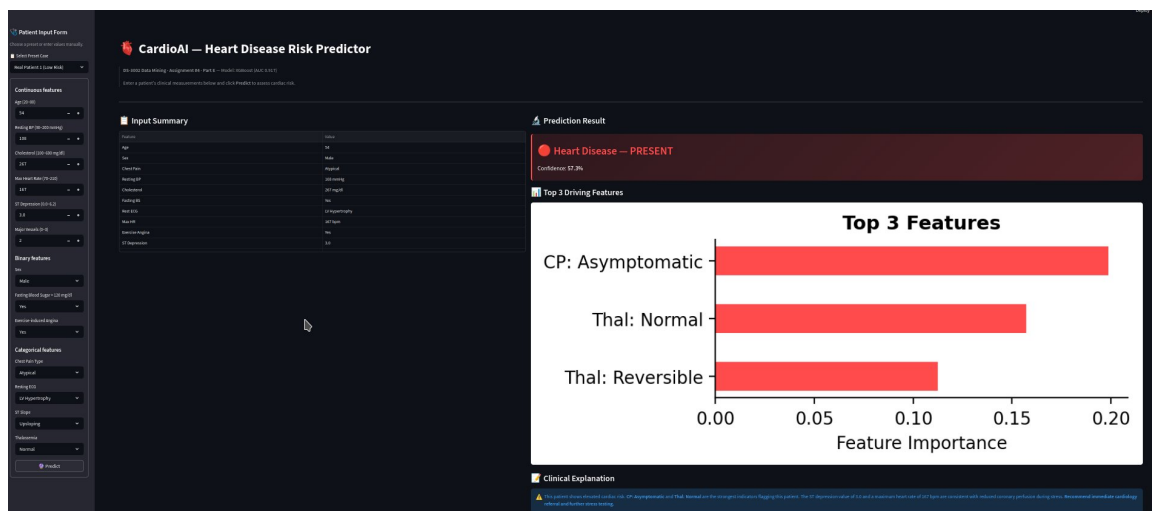


Figure 22: E2: Prediction Result (Disease Present) — Red risk banner showing “Heart Disease — PRESENT” at 57.3% confidence, Top 3 feature importance chart (red bars), and Clinical Explanation recommending immediate cardiology referral.

Clinical Explanation examples:

- *No Disease:* “This patient shows no strong indicators of heart disease. CP: Asymptomatic and Thal: Normal are the dominant model features, and both fall within

acceptable clinical ranges. Routine follow-up is advised; no urgent intervention required.”

- *Disease Present:* “This patient shows elevated cardiac risk. CP: Asymptomatic and Thal: Normal are the strongest indicators flagging this patient. The ST depression value of 3.0 and a maximum heart rate of 167 bpm are consistent with reduced coronary perfusion during stress. Recommend immediate cardiology referral and further stress testing.”

E3 — Submission Files

The following files are included in the submission ZIP:

File	Description
<code>app.py</code>	Streamlit dashboard source code
<code>best_xgb.pkl</code>	Saved XGBoost model (joblib)
<code>scaler.pkl</code>	Fitted StandardScaler (joblib)
<code>splits.pkl</code>	Train/test split arrays
<code>requirements.txt</code>	All Python dependencies

Table 15: E3: Submission File List

Conclusion

This report presented a complete machine learning pipeline on the UCI Cleveland heart disease dataset and the MNIST digit subset. Key findings are summarised below:

Part	Key Finding
Pre	297 rows retained after cleaning; 54/46 class balance; SMOTE applied on training only.
A1 (K-Means)	k=3 optimal; Cluster 1 is 91% disease-positive, a strong high-risk segment. ARI=0.25.
A2 (Hierarchical)	Ward linkage confirms 3 clusters; K-Means preferred (ARI 0.25 vs 0.19).
A3 (PCA/t-SNE)	10 components for 90% variance; t-SNE shows partial class overlap, confirming moderate task difficulty.
B1 (RF)	AUC=0.9247; oldpeak, thal, ca are top features; OOB stabilises at ≈ 50 trees.
B2 (XGBoost)	AUC=0.9174; mild overfitting detected at round 200; SHAP confirms clinically valid feature logic.
B3 (Comparison)	MLP has highest AUC (0.9375); RF recommended for deployment due to interpretability + speed.
C1 (SLP)	Linear model hits ceiling at $F1 \approx 0.754$; non-linear boundary needed.
C2 (MLP)	Medium architecture best; AUC=0.9375; Recall=0.8929; Early Stopping critical to avoid overfitting on small data.
C3 (Ablation)	Early Stopping is the single most important MLP component (loss explodes without it).
D1–D2 (CNN)	CNN reaches 97.75% accuracy vs 91.80% MLP baseline; surpasses baseline from epoch 1.
D3 (Viz)	First-layer filters detect edges and curves; feature maps confirm spatially-aware learning.

Table 16: Summary of Key Results Across All Parts

The heart disease dataset represents a genuinely difficult classification task — as evidenced by the t-SNE overlap and the moderate ARI scores in unsupervised analysis. No single model dominates across all metrics; the best strategy is a **layered deployment**: Random Forest as the primary interpretable engine, XGBoost with SHAP for detailed risk explanation, and the MLP as a secondary probability scorer. For the digit recognition module, the CNN is clearly superior and production-ready.

Every confusion matrix cell in the heart disease task represents a real patient. Models that a cardiologist can interrogate, explain, and audit are not merely preferable — in a clinical environment, they are the only ethically defensible choice.

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